



CUBAN NATIONAL CHEMISTRY CONTEST 2019

"Imagination is more important than knowledge."

Albert Einstein

Name(s): _____ Last name(s): _____

School: _____

Municipality: _____ Province: _____

CODE: _____ Test: **DAY 1. COMMON EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 1 problem set, which has 3 problems (problem 1, 20 items; problem 2, 4 items; problem 3, 2 items) and a total of 5 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem as its scoring (from a total of 100 points), as each item scoring (in marks) is indicated.

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Equation of ideal gas	$p \cdot V = n \cdot R \cdot T$
Gases constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta E + W$
Zero on Celsius scale	273,15 K	Gibbs' free energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Ionic product of wáter at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF ELEMENTS

1 1A																	18 8A	
1 H 1.008	2 2A											13 3A	14 4A	15 5A	16 6A	17 7A	2 He 4.003	
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18	
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95	
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80	
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3	
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)	
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)	

Problem 1. Miscellanea (20 points)

1.1	...	Total	Total
1	idem	20	20

Select the correct answer.

1.1. Chemical element with seven electrons in “s”-type orbitals:

A) Na	B) N	C) Fr	D) Cu
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1.2. Originate a yellow precipitate when reaction in aqueous solution:

A) AgNO_3 and K_2CrO_4	B) KI and $\text{Pb}(\text{NO}_3)_2$	C) CaCl_2 and NaOH	D) FeSO_4 and KOH
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1.3. Element with the highest secondary ionization energy:

A) Be	B) Mg	C) Ca	D) Sr
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1.4. Angular molecule:

A) NH_3	B) BeCl_2	C) SnCl_2	D) C_2H_2
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1.5. Diamagnetic species with bond order equal to 1:

A) O_2	B) O_2^{2-}	C) N_2	D) O_2^-
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1.6. The central atom has sp^3d hybridization:

A) SF_4	B) IF_5	C) CF_4	D) PF_3
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1.7. Element with the highest electron affinity:

A) S	B) P	C) Si	D) Al
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1.8. Substance with high melting and boiling points:

A) LiCl	B) PH_3	C) Na	D) HBr
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1.9. Largest ion:

A) K^+	B) Cl^-	C) Ca^{2+}	D) S^{2-}
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1.10. Gas denser than air:

A) CO	B) CH_4	C) H_2S	D) N_2
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1.11. Substance with the highest boiling point:

A) H_2O	B) HF	C) NH_3	D) HCl
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1.12. Apolar molecule:

A) IF	B) IF_3	C) IF_5	D) IF_7
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1.13. Glassware used to prepare solutions with exact concentration:

A) pipette	B) beaker	C) Erlenmeyer's flask	D) volumetric flask
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1.14. All the carbon atoms have sp^3 hybridization:

A) 2-butene	B) etoxyetane	C) butanone	D) ethyl ethanoate
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1.15. This species can be represented by three resonance structures in which formal charges can only take the values -1 , 0 , $+1$:

A) ClO_4^-	B) SO_3^{2-}	C) HCO_3^-	D) BrO_2^-
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1.16. Substance more soluble in water:

A) H_2S	B) $\text{Ca}(\text{OH})_2$	C) O_2	D) $\text{CH}_3\text{CH}_2\text{OH}$
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1.17. Element with green color when in the flame:

A) Ca	B) Cu	C) K	D) Pb
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1.18. Element impossible to keep in the solid-state during summer months in Cuba:

A) Sr	B) Ga	C) Na	D) Os
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1.19. Noble metal:

A) Mn	B) Rb	C) Zn	D) Pt
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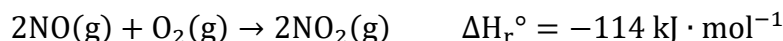
1.20. Redox reaction:

A) $\text{Na}_2\text{S}_2\text{O}_3(\text{ac}) + 2\text{HCl}(\text{ac}) = \text{S}(\text{s}) + \text{SO}_2(\text{g}) + \text{H}_2\text{O} + 2\text{NaCl}(\text{ac})$
B) $\text{CaCO}_3(\text{s}) + 2\text{HCl}(\text{ac}) = \text{CaCl}_2(\text{ac}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}$
C) $\text{NH}_3(\text{ac}) + \text{H}_2\text{SO}_4(\text{ac}) = \text{NH}_4\text{HSO}_4(\text{ac})$
D) $\text{K}_2\text{Cr}_2\text{O}_7(\text{ac}) + 2\text{KOH}(\text{ac}) = 2\text{K}_2\text{CrO}_4(\text{ac}) + \text{H}_2\text{O}$

Problem 2. Chemical kinetics (15 points)

2.1	2.2	2.3	2.4	Total	Total
12	7	3	8	30	15

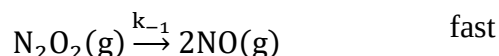
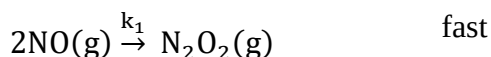
When a car releases nitrogen monoxide to the atmosphere, the following reaction happens.



2.1. Find the reaction rate law and calculate the value of the specific rate constant using the kinetic data below.

Experiment	Initial concentration ($\text{mol} \cdot \text{L}^{-1}$)		$-\text{dc}(\text{NO})/\text{dt}_{\text{initial}}$ ($\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$)
	NO	O ₂	
1	0,012	0,020	0,102
2	0,024	0,020	0,407
3	0,024	0,040	0,816

For the previous reaction, the following mechanism has been proposed.



2.2. Find an expression for the reaction rate as a function of the rate constant of the individual steps and the concentration of the reactants. Consider stationary conditions for the intermediaries.

2.3. ¿Under which experimental conditions the expression found by you in 2.2 is equivalent to the rate law from 2.1?

2.4. Draw the approximate form of an energy diagram *versus* reaction coordinate that matches the previously proposed mechanism. Indicate the enthalpy of the reaction.

Problem 3. Stoichiometry I (15 points)

3.1	3.2	Total	Total
15	15	30	15

A method to determine the empirical formula of an organic compounds is to carry out the complete combustion of a sample and analyze the resultant product mixture. The final result of the combustion of organic molecules with oxygen depends on the reaction conditions. In the presence of excess oxygen, the products are carbon dioxide and water; when the oxygen content is deficient or there are other atoms different than carbon or hydrogen, other products are formed. In the following two different situations in which this method is employed are described.

I- A 4,71 g sample of a solid organic substance **A** is introduced in a previously vacuumed container of 2,00 L capacity. In the container, oxygen was added until a pressure of 390,4 kPa was reached at 25 °C. After ignition, a black solid residual with mas 0,240 g was obtained. Additionally, 11,0 g of carbon dioxide and 2,70 g of water were formed. All the initial oxygen was consumed after the reaction.

3.1. Find the empirical formula of **A** if it contained carbon, hydrogen and oxygen. Neglect the water vapor pressure and the volume of the solids.

II- The complete combustion of a sample of alkanes **B** and **C** originated 61,6 g of carbon dioxide and 29,7 g of water. **B** is the higher homologous (it has an additional carbon in the molecular formula) of **C**. The molar ration between the amounts of **B** and **C** in the mixture is 3:2, respectively.

3.2. Find the molecular formula of **B** and **C**.